Fox School of Business
TEMPLE UNIVERSITY®

AS 2503

Homework 17 *(due: Nov. 12)* Fall 2026

Review of Lesson 17: One-Period Binomial Model

H 17-1 Let $S = \$100$, $K = \$95$, $r = 8\%$ compounded continuously, $T = 0.5$, $u = 1.3$, $d = 0.8$, and $n = 1$.

(a) Determine the option premium, Δ, and B for a European call.

(b) Determine the option premium, Δ, and B for a European put.

(c) Verify the consistency of your answers with the put-call parity.

(d) Suppose you observe a call price of \$17. What is your arbitrage strategy? Verify with a payout table.

(e) Suppose you observe a put price of \$8. What is your arbitrage strategy? Verify with a payout table.

Preparation for Lesson 18: Multi-Period Binomial Model

Watch these **video lectures** *and take proper notes:*

18a (14 mins), 18b (27 mins).

Now do the following exercise:

H 17-2 A stock index has a current price of 45 and pays no dividends. The index price evolves according to a binomial tree with period length 6 months, $u = 1.25$ and $d = 0.85$. The risk-free interest rate is 5%, compounded continuously. Determine the price of a 1-year European put option on this index with strike 47.